Skills Summary

Quadratics: Choosing a Method and Reading the Roots

Use this reference while completing your worksheet or when studying.

Decide before you solve. Put the equation in the form $ax^2 + bx + c = 0$ (or leave it as a perfect square), then:

- no x term, or the variable sits inside one square $\Rightarrow$ **square roots**
- factors over the integers (check a common factor first) $\Rightarrow$ **factoring**
- anything else, or you cannot see it quickly $\Rightarrow$ **quadratic formula** — it always works

Discriminant test: $\Delta = b^2 - 4ac$. A perfect-square Δ means it factors; $\Delta > 0$ two real roots, $\Delta = 0$ one, $\Delta < 0$ none.

1. Square roots (square-root-method)

$(x-h)^2 = k \;\Rightarrow\; x - h = \pm\sqrt{k} \;\Rightarrow\; x = h \pm \sqrt{k}$. Always write **both** signs. Divide off the outside coefficient first.

Example: Solve $(x-1)^2 = 25$.
Step 1. The variable appears only inside one square and there is no separate x term, so square roots is the shortest route.
Step 2. $x - 1 = \pm 5$, so $x = 1 + 5$ or $x = 1 - 5$: $\boldsymbol{x = 6}$ or $\boldsymbol{x = -4}$

Try it: Solve $2(x+4)^2 = 50$.

check: $\boldsymbol{x = 1}$ or $\boldsymbol{x = -9}$

2. Factoring (factoring-method)

Move everything to one side, factor, then use the **zero-product property**: if $AB = 0$ then $A = 0$ or $B = 0$.

For $x^2 + bx + c$: find two numbers with product c and sum b. Never divide both sides by x — that deletes the root $x = 0$.

Example: Solve $x^2 + 2x - 15 = 0$.
Step 1. It is already equal to zero and monic with small numbers, so hunt for a factor pair rather than reaching for the formula.
Step 2. Product -15, sum 2: use 5 and -3, so $(x+5)(x-3) = 0$ and $\boldsymbol{x = 3}$ or $\boldsymbol{x = -5}$

Try it: Solve $x^2 - 7x + 12 = 0$.

check: $\boldsymbol{x = 3}$ or $\boldsymbol{x = 4}$

3. Quadratic formula (quadratic-formula)

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Works on every quadratic. Use it the moment the factor hunt stalls — and simplify the radical instead of rounding, unless the problem asks for a decimal.

Example: Solve $x^2 - 6x + 4 = 0$.
Step 1. No integer pair multiplies to 4 and adds to -6, so factoring is a dead end and the formula is the efficient choice.
Step 2. $\Delta = 36 - 16 = 20$, so $x = \dfrac{6 \pm \sqrt{20}}{2} = \dfrac{6 \pm 2\sqrt{5}}{2}$, giving $\boldsymbol{x = 3 + \sqrt{5}}$ or $\boldsymbol{x = 3 - \sqrt{5}}$

Try it: Solve $x^2 + 2x - 4 = 0$.

check: $\boldsymbol{x = -1 + \sqrt{5}}$ or $\boldsymbol{x = -1 - \sqrt{5}}$

4. Reading roots in context (interpret-roots-in-context)

Solve the equation first, then ask what each root *means*. A root that solves the equation can still be impossible in the story — a negative length, a negative time, a fraction of a person. Say which root answers the question and why the other is rejected, and give the answer with its unit.

Example: A rectangle is 3 m longer than it is wide and has area 40 square metres. Find its width w.
Step 1. Area $= w(w+3) = 40$ becomes $w^2 + 3w - 40 = 0$; the numbers are small, so factor, then test each root against the picture.
Step 2. $(w+8)(w-5) = 0$ gives $\boldsymbol{w = 5}$ or $\boldsymbol{w = -8}$; a width cannot be negative, so the rectangle is **5** m wide

Try it: A ball thrown up from the ground has height $h = -16t^2 + 64t$ feet after t seconds. How long is it in the air?

check: $\boldsymbol{t = 4}$ seconds